SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – DEBUGGING & OPTIMIZATION TASK

Course Code:	CSA0305	**Course Title:**	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	**Assessment:**	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	**Total Marks:**	100
CO2 Statement:	*Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.*		
Student Name:	N.S.V.Trivikram	**Reg. No.:**	192525063
Section / Batch:	Aiml 2025	**Date:**	24/07/2026

Code of Conduct

IN.S.V.Trivikram (192525063) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: N.S.V.T.R

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25
Linked List Data Structure	List Operations	6 – 10	25
Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
		GRAND TOTAL:	**100 Marks**

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edge-case errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors
Code Quality & Readability	20	Clean, modular, well-documented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and well-justified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	No or incorrect explanation
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	No testing provided

Faculty In-charge	**Course Coordinator**	**Program Director**
Signature: ______________	Signature: ______________	Signature: ______________
Date: __________________	Date: __________________	Date: __________________

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack - Balancing Symbols 5 marks
- Q6 Singly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks
- Q10 Queue 5 marks

10/10 answered

Q1 Binary Search

5 marks

Identify the error in the following snippet.

```
int binarySearch(int arr[], int n, int key) {
int low = 0, high = n - 1;
while(low < high) {
int mid = (low + high) / 2;
if(arr[mid] == key)
return mid;
else if(arr[mid] < key)
low = mid;
else
high = mid;
}
return -1;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>

int binarySearch(int arr[], int n, int key) {
    int low = 0, high = n - 1;
    while (low <= high) {
        int mid = low + (high - low) / 2;
        if (arr[mid] == key)
            return mid;
        else if (arr[mid] < key)
            low = mid + 1;
        else
            high = mid - 1;
    }
    return -1;
}

int main() {
    int arr[] = {1, 2, 3, 4, 5, 6, 7, 8, 9};
    int n = sizeof(arr) / sizeof(arr[0]);
```

Test Input

stdin input (optional)

Output

```
Element found at index: 4
```

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← Back

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- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks
- Q10 Queue 5 marks

10/10 answered

Q2 Stack

5 marks

Find the error in the stack deletion operation. Justify it.

```
int pop() {
if(top == 0) {
printf("Underflow");
return -1;
}
return stack[top--];
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>

#define MAX 100

int stack[MAX];
int top = 0;

int pop() {
    if (top == 0) {
        printf("Underflow");
        return -1;
    }
    return stack[--top];
}

int main() {
    stack[top++] = 10;
    stack[top++] = 20;
    stack[top++] = 30;

    printf("%d\n", pop());
    printf("%d\n", pop());
    printf("%d\n", pop());
    printf("%d\n", pop());

    return 0;
}
```

Test Input

stdin input (optional)

Output

```
30
20
10
Underflow-1
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

Back

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- Q4 Stack 5 marks
- Q5 Stack - Balancing Symbols 5 marks
- Q6 Singly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks
- Q10 Queue 5 marks

10/10 answered

Q3 Stack — 5 marks

Check the snippet and identify the logical error.
int stack[5];
int top;
void init() {
top = 0;
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>

int stack[5];
int top;

void init() {
    top = -1;
}

void push(int value) {
    if (top == 4) {
        printf("Stack Overflow\n");
        return;
    }
    top = top + 1;
    stack[top] = value;
}

int pop() {
    if (top == -1) {
        printf("Stack Underflow\n");
        return -1;
    }
    int value = stack[top];
    top = top - 1;
    return value;
}

int main() {
```

Test Input

stdin input (optional)

Output

```
Popped: 30
Popped: 20
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

Back

QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack - Balancing Symbols 5 marks
- Q6 Singly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks
- Q10 Queue 5 marks

Q4 Stack — 5 marks

Identify the errors in the following snippet.
If(ch == ')') {
while(stack[top] != '(') {
postfix[j++] = pop();
}
pop();
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>

int pop(void);
extern int top;
extern char stack[];
extern char postfix[];
extern int j;
extern char ch;

void handle_close_parenthesis(void) {
    if (ch == ')') {
        while (top != -1 && stack[top] != '(') {
            postfix[j++] = pop();
        }
        if (top != -1 && stack[top] == '(') {
            pop();
        }
    }
}
```

Submitted

Test Input

stdin input (optional)

Output

```
Compilation Error:
C:/CCPPComp/CCPP3/bin/../lib/gcc/x86_64-w64-mingw32/10.3.0/../../../../x86_64-w64-mingw32/bin/ld.exe:
```

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10/10 answered

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
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- Q5 Stack - Balancing Symbols 5 marks
- Q6 Singly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks
- Q10 Queue 5 marks

10/10 answered

Q5 Stack - Balancing Symbols

5 marks

In balancing Bracket, you identify the error.
if((ch == ']' && stack[top] == '[') ||
(ch == ')' && stack[top] == '(')) {
pop();
} else {
return 0;
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>
#include <string.h>

#define MAX 100000

int isMatching(char open, char close) {
    return (open == '[' && close == ']') ||
           (open == '{' && close == '}') ||
           (open == '(' && close == ')');
}

int main() {
    char s[MAX];
    if (!fgets(s, sizeof(s), stdin)) return 0;


    size_t n = strlen(s);
    if (n > 0 && s[n - 1] == '\n') s[n - 1] = '\0';

    int top = -1;
    char stack[MAX];

    for (size_t i = 0; s[i] != '\0'; i++) {
        char ch = s[i];

        if (ch == '[' || ch == '{' || ch == '(') {
            stack[++top] = ch;
        } else if (ch == ']' || ch == '}' || ch == ')') {
```

Test Input

stdin input (optional)

Output

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack - Balancing Symbols 5 marks
- Q6 Singly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks
- Q10 Queue 5 marks

10/10 answered

Q6 Singly Linked List

5 marks

What is the issue?
temp = head
while temp.next != NULL:
print(temp.data)
temp = temp.next

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>
#include <stdlib.h>

struct Node {
    int data;
    struct Node* next;
};

void print_list(struct Node* head) {
    struct Node* temp = head;
    while (temp != NULL) {
        printf("%d\n", temp->data);
        temp = temp->next;
    }
}

int main() {
    struct Node* head = (struct Node*)malloc(sizeof(struct Node));
    struct Node* second = (struct Node*)malloc(sizeof(struct Node));
    struct Node* third = (struct Node*)malloc(sizeof(struct Node));

    head->data = 1;
    head->next = second;

    second->data = 2;
    second->next = third;

    third->data = 3;
    third->next = NULL;

    print_list(head);
```

Test Input

stdin input (optional)

Output

```
1
2
3
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

Back

QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack - Balancing Symbols 5 marks
- Q6 Singly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks
- Q10 Queue 5 marks

10/10 answered

Q7 Stack

5 marks

Point out the error.
POP(stack):
If top == 0:
print("Underflow")
else:
return stack[top--]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>

#define MAX 100

int stack[MAX];
int top = -1;

void push(int x) {
    if (top == MAX - 1) {
        printf("Overflow\n");
    } else {
        stack[++top] = x;
    }
}

int pop() {
    if (top == -1) {
        printf("Underflow\n");
        return -1;
    } else {
        return stack[top--];
    }
}

int main() {
    int n, i, x;

    printf("Enter number of elements to push: ");
    scanf("%d", &n);
```

Test Input

```
10
20
30
```

Output

```
Enter number of elements to
push: Enter 10 elements:
How many times to pop? 30
30
20
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

Back

QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack - Balancing Symbols 5 marks
- Q6 Singly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks
- Q10 Queue 5 marks

10/10 answered

Q8 Linked List

5 marks

Identify the errors in the following code
while current != NULL:
current.next = prev
prev = current
current = current.next

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>
#include <stdlib.h>

struct Node {
    int data;
    struct Node* next;
};

struct Node* reverseList(struct Node* head) {
    struct Node* current = head;
    struct Node* prev = NULL;
    struct Node* nextNode = NULL;

    while (current != NULL) {
        nextNode = current->next;
        current->next = prev;
        prev = current;
        current = nextNode;
    }

    return prev;
}

void printList(struct Node* head) {
    struct Node* temp = head;
    while (temp != NULL) {
        printf("%d -> ", temp->data);
        temp = temp->next;
    }
    printf("NULL\n");
}
```

Test Input

```
1
2
3
4
```

Output

```
Original list: 1 -> 2 -> 3 -> 4 ->
NULL
Reversed list: 4 -> 3 -> 2 -> 1 ->
NULL
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack · Balancing Symbols 5 marks
- Q6 Singly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks
- Q10 Queue 5 marks

10/10 answered

Q9 Linked List

5 marks

What is the error in the below code?
new.next = head
head = new

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>
#include <stdlib.h>

struct Node {
    int data;
    struct Node* next;
};

struct Node* head = NULL;

void insertAtHead(int value) {
    struct Node* new = (struct Node*)malloc(sizeof(struct Node));
    new->data = value;
    new->next = head;
    head = new;
}

void printList() {
    struct Node* temp = head;
    while (temp != NULL) {
        printf("%d ", temp->data);
        temp = temp->next;
    }
    printf("\n");
}

int main() {
    int n, value;

    scanf("%d", &n);
    for (int i = 0; i < n; i++) {
        scanf("%d", &value);
        insertAtHead(value);
```

Test Input

```
5
10
20
30
```

Output

```
50 40 30 20 10
```

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192525063RA

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue · Deque 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks
- Q10 Stack · Postfix Expression 5 marks

10/10 answered

Q2 Circular Queue

5 marks

Identify the issue in the following snippet for Circular Queue in Traffic Signal.
rear = (rear + 1) % MAX;
queue[rear] = vehicle;
if(rear == front){
printf("Queue Full");
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>
#include <stdbool.h>

#define MAX 5

int queue[MAX];
int front = -1, rear = -1;


bool isFull() {
    return ((rear + 1) % MAX) == front;
}

bool isEmpty() {
    return front == -1;
}


void enqueue(int vehicle) {
    if (isFull()) {
        printf("Queue Full!\n");
        return;
    }

    if (isEmpty()) {
        front = rear = 0;
    } else {
        rear = (rear + 1) % MAX;
    }
```

Test Input

```
1
101
3
2
```

Output

```
=== Traffic Signal Circular Queue ===
1. Enqueue vehicle
2. Dequeue vehicle
3. Display queue
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue - Deque 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks
- Q10 Stack - Postfix Expression 5 marks

10/10 answered

Q3 Stack

5 marks

Point out the error.
if top == MAX:
Overflow

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>
#include <stdlib.h>

#define MAX 5

struct Stack {
    int items[MAX];
    int top;
};


void initStack(struct Stack *s) {
    s->top = -1;
}


int isFull(struct Stack *s) {

    if (s->top == MAX - 1) {
        return 1;
    } else {
        return 0;
    }
}

int isEmpty(struct Stack *s) {
    if (s->top == -1) {
        return 1;
    } else {
        return 0;
    }
}

```

Test Input

```
1
10
1
20
```

Output

```
Stack Operations:
1. Push
2. Exit

Enter choice: Enter value to
push: Pushed: 10
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue - Deque 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks
- Q10 Stack - Postfix Expression 5 marks

10/10 answered

Q4 Stack

5 marks

Identify the issue in the below pseudocode.
POP():
if top == 0 → Underflow

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>
#include <stdlib.h>

#define MAX 100

typedef struct {
    int arr[MAX];
    int top;
} Stack;

void initStack(Stack *s) {
    s->top = -1;
}

int isEmpty(Stack *s) {
    return s->top == -1;
}

int pop(Stack *s, int *out) {
    if (isEmpty(s)) {
        return 0;
    }
    *out = s->arr[s->top];
    s->top--;
    return 1;
}

int main() {
    Stack s;
    initStack(&s);

    int val;
    if (pop(&s, &val)) {
```

Test Input

```
20
30
45
55
```

Output

```
Underflow
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue - Deque 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks
- Q10 Stack - Postfix Expression 5 marks

10/10 answered

Q5 Doubly Linked List

5 marks

Identify the error in this Doubly Linked List node deletion.
void deleteNode(Node* del) {
if (del->prev != NULL)
del->prev->next = del->next;
if (del->next != NULL)
del->next->prev = del->prev;
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
#include <stdlib.h>

typedef struct Node {
    int data;
    struct Node *prev;
    struct Node *next;
} Node;

Node* createNode(int val) {
    Node* n = (Node*)malloc(sizeof(Node));
    n->data = val;
    n->prev = NULL;
    n->next = NULL;
    return n;
}

void deleteNode(Node* del) {
    if (del == NULL) return;

    // Link previous node to next node
    if (del->prev != NULL)
        del->prev->next = del->next;

    if (del->next != NULL)
        del->next->prev = del->prev;


```

Test Input

```
10
20
30
40
```

Output

```
10 30 40
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
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- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue - Deque 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks
- Q10 Stack - Postfix Expression 5 marks

10/10 answered

Q6 Queue

5 marks

Identify the error in the Queue enqueue logic.
void enqueue(int data) {
if (rear == MAX) return;
queue[rear++] = data;
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
#include <stdlib.h>

#define MAX 100

int queueArr[MAX];
int front = -1;
int rear  = -1;

int isFull() {
    return (rear == MAX - 1);
}

int isEmpty() {
    return (front == -1);
}


void enqueue(int data) {
    if (isFull()) {
        printf("Queue Overflow\n");
        return;
    }
    if (isEmpty()) {
        front = 0;
    }
    queueArr[++rear] = data;
}

int dequeue() {
```

Test Input

```
10
20
```

Output

```
10
20
```

Pending manual review

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue - Deque 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks
- Q10 Stack - Postfix Expression 5 marks

10/10 answered

Q7 Queue - Deque

5 marks

Find the error in this Deque insertFront operation.
void insertFront(int data) {
if (front == 0) front = MAX - 1;
else front = front - 1;
deque[front] = data;
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>
#define MAX 100

int deque[MAX];
int front = -1, rear = -1;

int isFull() {
    return (front == 0 && rear == MAX - 1) || (front == rear + 1);
}

int isEmpty() {
    return front == -1;
}

// Correct insertFront for circular deque
void insertFront(int data) {
    if (isFull()) {
        printf("Deque Overflow\n");
        return;
    }

    if (isEmpty()) {
        front = rear = 0;
    } else if (front == 0) {
        front = MAX - 1;
    } else {
        front = front - 1;
    }

```

Test Input

```
20
10
5
```

Output

```
Deque elements: 20 10 5
Deleted Front: 20
Deleted Rear: 5
Deque elements: 10
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue - Deque 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks
- Q10 Stack - Postfix Expression 5 marks

10/10 answered

Q8 Binary Search

5 marks

Identify the bug in this Recursive Binary Search.
int binarySearch(int arr[], int l, int r, int x) {
int mid = l + (r - l) / 2;
if (arr[mid] == x) return mid;
if (arr[mid] > x) return binarySearch(arr, l, mid, x);
return binarySearch(arr, mid, r, x);
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>


int binarySearch(int arr[], int l, int r, int x) {

    if (r >= l) {
        int mid = l + (r - l) / 2;


        if (arr[mid] == x) {
            return mid;
        }


        if (arr[mid] > x) {
            return binarySearch(arr, l, mid - 1, x);
        }


        return binarySearch(arr, mid + 1, r, x);
    }


    return -1;
}

int main() {
    int arr[] = {2, 3, 4, 10, 40};
```

Test Input

```
stdin input (optional)
```

Output

```
Element is present at index 3
```

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← Back

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- Q9 Circular Queue 5 marks
- Q10 Stack - Postfix Expression 5 marks

10/10 answered

Q10 Stack - Postfix Expression

5 marks

Find the bug in the evaluation of a Postfix expression.

```
while (expression[i] != '\0') {
if (isdigit(expression[i])) push(expression[i] - '0');
else {
int op1 = pop();
int op2 = pop();
// perform operation: result = op1 + op2;
push(result);
}
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
#include <ctype.h>
#include <stdlib.h>

#define MAX 100

int stackArr[MAX];
int top = -1;

void push(int x) {
    if (top >= MAX - 1) {
        printf("Stack Overflow\n");
        exit(1);
    }
    stackArr[++top] = x;
}

int popVal() {
    if (top < 0) {
        printf("Stack Underflow\n");
        exit(1);
    }
    return stackArr[top--];
}
```

Test Input

```
23*55*+9-
```

Output

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

- Q1 Stack 5 marks
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- Q9 Circular Queue 5 marks
- Q10 Stack - Postfix Expression 5 marks

10/10 answered

Q1 Stack

5 marks

Point out the error?

PEEK(stack):

return stack[top+1]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>

#define MAX 100

int stack[MAX];
int top = -1;

void push(int x) {
    if (top == MAX - 1) {
        printf("Stack Overflow!\n");
        return;
    }
    stack[++top] = x;
}

int peek() {
    if (top == -1) {
        printf("Stack Underflow! (Empty Stack)\n");
        return -1;
    }

    return stack[top];
}

int main() {
    int n, x;

    printf("Enter number of elements to push into stack: ");
    scanf("%d", &n);

    printf("Enter %d elements:\n", n);
    for (int i = 0; i < n; i++) {
```

Test Input

```
5
10
20
30
```

Output

```
Enter number of elements to push into stack: Enter 5 elements:
Top element (PEEK) = 55
```

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← Back

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- Q1 Stack 5 marks
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- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks
- Q10 Stack - Postfix Expression 5 marks

10/10 answered

Q9 **Circular Queue** 5 marks

Identify the bug in this Circular Queue dequeue logic.

```
int dequeue() {
if (front == -1) return -1;
int data = queue[front];
front = (front + 1) % MAX;
return data;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>

#define MAX 5

int queue[MAX];
int front = -1;
int rear = -1;

void enqueue(int value) {
    if ((rear + 1) % MAX == front) {
        return;
    }
    if (front == -1) front = 0;
    rear = (rear + 1) % MAX;
    queue[rear] = value;
}

int dequeue() {
    if (front == -1) {
        return -1;
    }

    int data = queue[front];

    if (front == rear) {
        front = -1;
        rear = -1;
    } else {
```

Test Input

```
10
20
30
```

Output

```
10
20
20
-1
```

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